

Tema Laborator 3

$$1. \quad X \sim \begin{pmatrix} 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 \\ \frac{1}{36} & \frac{2}{36} & \frac{3}{36} & \frac{4}{36} & \frac{5}{36} & \frac{6}{36} & \frac{5}{36} & \frac{4}{36} & \frac{3}{36} & \frac{2}{36} & \frac{1}{36} \end{pmatrix}$$

a) Pentru găsirea probabilității evenimentului ca suma 5 să apară înainte de suma 7, încercăm cu probă la a n-a aruncare nu s-a nimerit nici 5 nici 7 iar la a n-a aruncare s-a nimerit 5, not cu E_n .

fie:

$A_i = \{ \text{la aruncarea } i \text{ suma zarurilor este } 5 \}$

$B_i = \{ \text{la aruncarea } i \text{ suma zarurilor este } 7 \}$

$$P(E_n) = P((\overline{A_1} \cap \overline{B_1}) \cap (\overline{A_2} \cap \overline{B_2}) \cap \dots \cap (\overline{A_{n-1}} \cap \overline{B_{n-1}}) \cap A_n)$$

dar toate A_i și B_i sunt egale, cum aruncările sunt

independente

$$P(E_n) = P((\overline{A_1} \cap \overline{B_1})^{n-1} \cap A_n) = \left(\frac{26}{36}\right)^{n-1} \cdot \frac{4}{36}$$

fie

$C = \{ \text{suma 5 apare înainte de suma 7} \}$

$$P(C) = P\left(\bigcup_{n=1}^{\infty} E_n\right)$$

$$= \sum_{n=1}^{\infty} P(E_n)$$

$$= \frac{1}{9} \sum_{n=1}^{\infty} \left(\frac{13}{18}\right)^{n-1}$$

$$= \frac{1}{9} \cdot \frac{1}{1 - \frac{13}{18}}$$

$$= \frac{1}{9} \cdot \frac{18}{5} = \frac{2}{5}$$

- b) Pentru înlocuirea sumei de 5 cu 2 obs că singura diferență este că este doar o posibilitate din 36 de a pica 2.

fie:

$D_i = \{ \text{la aruncarea } i \text{ suma zarurilor este } 2 \}$

$F_n = \text{în primele } n-1 \text{ aruncări nu a apărut nici suma } 2, \text{ nici } 7, \text{ iar la aruncarea } n \text{ a apărut suma } 2.$

$$\text{analog: } P(F_n) = P((\overline{C_1} \cap \overline{D_1})^{n-1} \cdot D_1) = \left(\frac{29}{36}\right)^{n-1} \cdot \frac{1}{36}$$

$G = \text{suma } 2 \text{ apare înainte de suma } 7$

$$P(G) = P\left(\bigcup_{n=1}^{\infty} F_n\right)$$

$$= \frac{1}{36} \sum_{n=1}^{\infty} \left(\frac{29}{36}\right)^{n-1}$$

$$= \frac{1}{36} \sum_{n=0}^{\infty} \left(\frac{29}{36}\right)^n$$

$$= \frac{1}{36} \cdot \frac{1}{1 - \frac{29}{36}} = \frac{1}{36} \cdot \frac{36}{7} = \frac{1}{7}$$

$$2. \quad X \sim \begin{pmatrix} -1 & 0 & 1 \\ 0,3 & 0,2 & 0,5 \end{pmatrix}$$

$$P\left(X > -\frac{1}{3}\right) = P(\{X=0\} \cup \{X=1\}) = 0,2 + 0,5 = 0,7$$

$$P\left(X < \frac{1}{4} \mid X \geq -\frac{1}{2}\right) = \frac{P\left(-\frac{1}{2} \leq X < \frac{1}{4}\right)}{P\left(X \geq -\frac{1}{2}\right)} = \frac{0,2}{0,2+0,5} = \frac{0,2}{0,7} = \frac{2}{7}$$

3.

X - variabilă aleatoare cu val în $\mathbb{N}$, aî $p_n = P(X=n), n \in \mathbb{N}$
nt a dem că $i \Leftrightarrow ii$ dem că $i \Rightarrow ii$ și că $ii \Rightarrow i$

a)

$n \rightarrow \infty$

X - var Poisson de param $\lambda \Rightarrow p_n = e^{-\lambda} \frac{\lambda^n}{n!}$

$$\frac{p_n}{p_{n-1}} = \frac{e^{-\lambda} \frac{\lambda^n}{n!}}{e^{-\lambda} \frac{\lambda^{n-1}}{(n-1)!}} = \frac{\lambda}{n}$$

$$\frac{p_n}{p_{n-1}} = \frac{\lambda}{n} \Rightarrow$$

$$\frac{p_n}{p_0} = \frac{p_n}{p_{n-1}} \cdot \frac{p_{n-1}}{p_{n-2}} \cdot \dots \cdot \frac{p_1}{p_0} = \frac{\lambda^n}{n \cdot (n-1) \cdot \dots \cdot 1} = \frac{\lambda^n}{n!} \Rightarrow$$

$$p_n = p_0 \frac{\lambda^n}{n!}$$

$$p_0 = e^{-\lambda} \left(\text{din } \sum_{h=0}^{\infty} p_h = 1 \right) \Rightarrow p_n = e^{-\lambda} \cdot \frac{\lambda^n}{n!}$$

h) ~~unalog dem cu i) cu ii) cu ii) cu i)~~

$$X \sim P(\lambda)$$

i) $h = ? \Rightarrow p(X=h)$ max

stim de la a) cu $\frac{p_n}{p_{n-1}} = \frac{\lambda}{n}$, la raportul

$$p(X=n) \geq p(X=n-1), \text{ de } \lambda \geq n \quad \left\{ \Rightarrow \right.$$

$$p(X=n) < p(X=n-1), \text{ de } \lambda < n$$

$$\Rightarrow \text{at } h = [\lambda] \text{ este max } \Rightarrow p(X=[\lambda]) = e^{-\lambda} \cdot \frac{\lambda^{[\lambda]}}{[\lambda]!} \text{ val max}$$

ii) de la a) avem cu $\frac{p_n}{p_{n-1}} = \frac{\lambda}{n}$, at $n > 0$

observam cu maximal este atins pentru $X = n$

$$f(x) = \begin{cases} \alpha x^2 e^{-hx}, & x \geq 0, h > 0 \\ 0, & x < 0 \end{cases}$$

a) stim cu $\int_{-\infty}^{+\infty} f(x) dx = 1 \Leftrightarrow$

$$\int_{-\infty}^0 0 dx + \int_0^{+\infty} \alpha x^2 e^{-hx} dx = 1 \quad \begin{matrix} y = hx \\ dy = h dx \end{matrix}$$

$$0 + \frac{\alpha}{h^3} \int_0^{+\infty} y^2 e^{-y} dy = \frac{2\alpha}{h^3} = 1 \Rightarrow 2\alpha = h^3 \Rightarrow \alpha = \frac{h^3}{2}$$

$$b) f(t) = \begin{cases} \frac{h^3}{2} \cancel{t^2} t^2 e^{-ht}, & t \geq 0 \\ 0, & t < 0 \end{cases}, h > 0$$

$$F(t) = \int_{-\infty}^t f(t) dt$$

$$= \int_{-\infty}^0 0 dt + \int_0^t \frac{h^3}{2} \cdot t^2 e^{-ht} dt =$$

$$\Rightarrow F(t) = \begin{cases} 0, & t \leq 0 \\ 1 - \frac{h^2 t^2 + 2ht + 2}{2} \cdot e^{-ht}, & t \in (0, +\infty) \\ 1, & t > 1 \end{cases}$$

$$c) P(0 < X < h^{-1}) = P\left(0 < X < \frac{1}{h}\right) = F\left(\frac{1}{h}\right) - F(0)$$

$$= 1 - \frac{1 + 2 + 2}{2} \cdot e^{-1}$$

$$= 1 - \frac{5}{2e}$$

5. X - var discretă

$$P(X=h) = \begin{cases} \frac{(1-p)^h}{-h \log(p)}, & h \geq 1 \\ 0, & h = 0 \end{cases}, 0 < p < 1$$

$$a) E[X] = \sum_{h=0}^{+\infty} h P(X=h) = \sum_{h=1}^{\infty} h \frac{(1-p)^h}{-h \log(p)} = -\frac{1}{\log(p)} \sum_{h=1}^{\infty} (1-p)^h$$

$$= -\frac{1}{\log p} \left[\frac{1}{1-(1-p)} - 1 \right]$$

$$= -\frac{1-p}{p \log(p)}$$

$$\begin{aligned}
 b) \quad E[X^2] &= \sum_{h=0}^{\infty} h^2 p(X=h) = \sum_{h=1}^{\infty} h^2 \frac{(1-p)^h}{-h \log p} \\
 &= -\frac{1}{\log(p)} \sum_{h=1}^{\infty} h (1-p)^h = -\frac{1}{\log(p)} (1-p) \sum_{h=1}^{\infty} ((1-p)^{h-1}) \\
 &= -\frac{1}{\log(p)} \cdot \frac{(1-p)}{p^2} = -\frac{1-p}{p^2 \log(p)}
 \end{aligned}$$

$$\begin{aligned}
 c) \quad \text{Var}[X] &= E[X^2] - E[X]^2 \\
 &= -\frac{1-p}{p^2 \log(p)} - \left(-\frac{1-p}{p \log(p)} \right)^2 \\
 &= -\frac{\log(p)}{1-p} - \frac{(1-p)^2}{p^2 \log^2(p)} \\
 &= -\frac{(1-p)(1-p + \log(p))}{p^2 \log^2(p)}
 \end{aligned}$$

b. a) X - var al cu val în $\mathbb{N}$

$$E[X] \neq \sum_{n \geq 1} p(X \geq n)$$

$$\begin{aligned}
 p(X > n) &= p(X = n+1) + p(X = n+2) + \dots \\
 &= \sum_{h=n+1}^{\infty} p(X=h)
 \end{aligned}$$

$$\sum_{n=0}^{\infty} p(X > n) = \sum_{n=0}^{\infty} \sum_{h=n+1}^{\infty} p(X=h) =$$

$$= (p(X=1) + p(X=2) + \dots) + (p(X=2) + p(X=3) + \dots) + \dots$$

$$= 1 \cdot p(X=1) + 2 \cdot p(X=2) + \dots + n \cdot p(X=n) + \dots$$

$$= \sum_{h=1}^{\infty} h p(X=h) = \sum_{h=0}^{\infty} h p(X=h) = E[X]$$

6) x - var al cu val in $\mathbb{R}_+$

$$E[x] = \int_0^{+\infty} p(x \geq t) dt$$

$$\int_0^{+\infty} p(x \geq t) dt = \int_0^{+\infty} E[1_{\{x \geq t\}}] dt \quad \text{Tonelli - Fubini} \Rightarrow$$

$$= E \left[\int_0^{+\infty} 1_{\{x \geq t\}} dt \right] = E \left[\int_0^x dt \right] = E \left[x \Big|_0^x \right] =$$

$$= E[(x - 0)] = E[x]$$

7. a) x - var repartizată exp (de rezervă α
 pe o pereche liniei de memorie:

$$P(x > s+t | x > s) = P(x > t)$$

deci $x \sim \text{exponen}(\alpha)$

$$f(t) = \alpha e^{-\alpha t} 1_{(0, +\infty)}(t)$$

$$F(t) = (1 - e^{-\alpha t}) 1_{(0, +\infty)}(t)$$

$$P(x > t) = 1 - P(x \leq t)$$

$$= 1 - F(t)$$

$$= 1 - (1 - e^{-\alpha t})$$

$$= e^{-\alpha t} \Rightarrow$$

$$P(x > s+t | x > s) = \frac{P(x > s+t, x > s)}{P(x > s)}$$

$$= \frac{P(x > s+t)}{P(x > s)} = \frac{e^{-\alpha(s+t)}}{e^{-\alpha s}}$$

$$= \frac{e^{-\alpha s} \cdot e^{-\alpha t}}{e^{-\alpha s}} = e^{-\alpha t} = P(x > t)$$

b) din rel (1) $\Rightarrow P(X > s+t) = P(X > s) \cdot P(X > t)$

notăm $f(x) = P(X > x)$

avem că $f(s+t) = f(s)f(t)$ cu $s > 0, t > 0$

Pentru a dem că X - var al este distr exp, trebuie să
 ne folosim de funcția Gaudy

pentru $s=t \Rightarrow f(s+s) = f(s)f(s)$

$$f(2s) = f^2(s)$$

$$f(2s+s) = f(2s)f(s)$$

$$f(3s) = f^3(s)$$

$\vdots$ prin ind mat

$$f(hs) = f^h(s) \text{ cu } h \in \mathbb{N} \quad (2)$$

dând val $s = \frac{1}{2} \Rightarrow f\left(\frac{1}{2} + \frac{1}{2}\right) = f^2\left(\frac{1}{2}\right) \Rightarrow f(1) = f^2\left(\frac{1}{2}\right) \Rightarrow$

$$f\left(\frac{1}{2}\right) = f^{\frac{1}{2}}(1)$$

$\vdots$ prin ind mat

$$f\left(\frac{1}{h}\right) = f^{\frac{1}{h}}(1) \quad (3)$$

$\xrightarrow{(2), (3)} f\left(\frac{m}{n}\right) = f\left(m \cdot \frac{1}{n}\right) = f^m\left(\frac{1}{n}\right) = f^{\frac{m}{n}}(1) \Rightarrow$

$$\Rightarrow f(q) = f^q(1) \quad (\forall) q \in \mathbb{Q}_+$$

fiind $r \in \mathbb{R}_+ \setminus \mathbb{Q}_+ \Rightarrow \exists (q_n)_{n \in \mathbb{N}}$ a.i. $q_n \rightarrow r \Rightarrow$

$$\Rightarrow f(q_n) \rightarrow f(r) \Rightarrow f(r) = f^r(1) \Rightarrow$$

$$f(x) = e^{-x \log \frac{1}{f(1)}} \Rightarrow X - \text{var al repart exp}$$